Analysis and Implementation of A Platform for Viewing 3-D Figures in General Formats

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Abstract: Currently, 3-D figures have not yet become the mainstream format for viewing figures on the Web. One important reason behind this phenomenon is that there is no generic standard for the broad variety of formats of 3-D figures. In this paper, we analyze different methods for displaying 3-D model files in the browser, and develop a small platform that enables previewing of multiple types of 3-D files in the same browser. File transformation and loading time is tested. Experimental results show that the proposed application can support effective online viewing due to its ability to read different types of files across Windows and MAC platforms. The proposed application can be inserted into the Web as a plug-in to display various 3-D figures. Only the source files in the format of 3-D files are needed to apply it extensively to online shopping, virtual reality and other Web applications.
Key words: 3-D files; online previewing; file format conversion; Three.js

1. **Introduction**

Traditionally, the Web provides users with various multimedia information in the form of pictures, words and animations. However, this form of information display does not support sufficient interaction with users. The 3-D pictures can represent scenarios in a more lifelike and interactive manner.

Due to limits on the online virtual environment in e-commerce, the buyers can only obtain information on the commodities using pictures, text descriptions or customer services, rather than touch the actual commodities by themselves. Hence, the buyers are usually skeptical about descriptions of the commodities on the Web. And it is usually reported that the commodities are not as good as the pictures describe. In this context, the customers expect the commodities to be displayed more vividly. Image how cool it will be to showcase all commodities in Taobao using 3-D models. But 3-D viewing has not gained widespread popularity. It is not recognized as a mainstream means of information display, due to the following reasons.

(1) 3-D models occupy much more disk space than the traditional pictures. It takes more time to transmit the data of 3-D models if the network bandwidth is low, affecting user experience of viewing information on the Web. One method that allows the 3-D pictures to be transmitted as widely as the traditional pictures is to compress the data of 3-D models. The authors in [18] proposed a method to greatly compress the size of 3-D files using a triangular grid. Their work formed foundation for widespread transmission of 3-D pictures. Furthermore, displaying 3-D contents on the Web is also enabled by advances in

the Internet bandwidth, computer graphics processing and Web development technologies.

(2) Formats of 3-D graphics are too many. As we all know, the formats of 2-D graphics, such as png, jpeg and gif, are fixed and mature. As early as in 1992, the Joint Graphics Group of International Organization Organization (ISO) has formulated the format of JPEG files and this greatly stimulated the transmission of 2-D graphics across the Internet. However, there are many formats of 3-D graphics at present, such as step, dwg and stl, which assume dominance in their respective areas. For example, dwg is the format that Autodesk uses by default in Auto CAD and it is widely used for modeling. step is the format that was proposed by ISO and that has been widely used in the shipping, manufacturing and other light industries. stl is the format used by computer graphics applications to represent triangular grids. It merely stores the geometrical information of the triangular slices and thus is very simple and widely used in the Internet online applications. It can be seen that different areas have different mainstream formats of 3-D models. But different types of 3-D files cannot be accessed in the same software, severely affecting the exchange of information between different areas.

To address the problem of isolation, we propose a novel technology to view different types of 3-D models in general applications and develop a small-scale platform to view the files of 3-D models online. Experiments are performed to evaluate its performance. Results show that the proposed technology is able to read and display different types of high-accuracy 3-D models across different platforms. Our work makes theoretical and technological contributions to online viewing of 3-D graphics.

2. 3-D Framework Analysis

In order to display the 3D images online, we need to analyze existing web3D techniques and adopt appropriate frameworks. Web3D refers to the technology of displaying 3D graphics on the Web browser. It models the 3D files in real time and displays 3D graphics using the rendering engine. Many world-famous companies around the world have formulated their own Web3D-based 3D graphics displaying technologies and solutions to display 3D graphics as expected [11].

Table 1 Cores of Web3D and comparison of their features

:echnologies	Principles	Features	Limitations
.-based technology	The server provides VRML files and resources. The browser parses and analyzes the files using the plug-in.	It is implemented via programming, visualization and 3D modeling. The files have a large size of data.	It needs to install plug-in beforehand and is poorly compatible with other technologies.
)ased technology	It directly executes the program via the browser and directly	It is implemented through programming and	It makes great demands on CPU calculation ability.

	displays the graphics after rendering.	3D modeling. The files have a small size of data.	
ning transmission	It directly embeds the interactive virtual scenarios and objects into the video.	It is implemented by combining the real-life graphics with virtual scenarios.	Real-life graphics need to be prepared.

Although many solutions are available currently to display 3D graphics on the Web, they have some limitations.

(1) The VRML technology needs to install the plug-in into the client. That is, the client of the user must download and install additional toolkit in order to display 3D graphics using this technology. If the plug-in is not installed correctly, all 3D graphics cannot be displayed on the browser. For users who are not adept in the use of computer, they have a lot of trouble installing special plug-ins.

(2) The Java-based technology needs to proves a large amount of data and makes great demands on CPU calculation ability.

(3) The streaming transmission-based technology requires real-life graphics. This need for graphics on real objects and scenarios blocks further development of it.

To address these problems, Three.js is used in this paper as the framework to view 3D graphics on the Web. As a WebGL-based third-party library, Three.js takes SML as its core and is being put into use in practice. It is also capable of perfectly incorporating 3D graphics into the Web pages. With this framework, we can quickly construct 3D scenarios, including cameras, illuminations and other objects. The WebGL-based Three.js engine has the following advantages [15].

(a) No need for extra plug-in. Three.js directly supports GPU hardware acceleration for 3D graphics. Hence, all users can directly use this technology after opening the browser.

(b) Open-source: Three.js follows the MIT open-source protocol and allows to be copied, distributed and modified freely. Hence, the cost is very low.

(c) Platform independence: Three.js is developed using JavaScript, and all browsers have built-in JavaScript engines. Hence, Three.js can be used on almost all browsers.

(d) Ease of use: Three.js has implemented many 3D modeling functions and packaged important methods and rendering cycles required to render various 3D scenarios. Hence, even programmers without a good command of computer graphics can use this technology effortlessly [9]. This facilitates the promotion of Three.js.

By combining with the traditional Web technologies, the WebGL-based Three.js framework can read data from the source file and perfectly display 3D graphics [10]. This solution is not only platform independent but contributes to subsequent development.

3. Principles of File Browsing Technologies

Due to the dramatic difference between the standard for displaying 3D graphics on the Web and the format of the 3D source file, it is very complicated to convert the source file into the graphics that can be displayed on the browser. In order to achieve independence of

platform and file format, we need to prove that irrespective of changes of file formats, the core elements inside are still the representations of its basic geometric solids. That is, as long as the key information of the basic geometric solids are exported, the geometric solids can be reconstructed using these elements. Hence, we can display the geometric solids on the Web by importing their elements into the Web.

Any 3D model consists of vertexes, edges and surfaces. Geometric and topological data are indispensable to a complete representation of geometric solids [7].

Geometrical data:

It is able to determine the location and size of 3D models in the Euclidean space, such as the coordinates of vertexes, and normal vector of the edge's directional vector. Points, edges and surfaces can be transformed to one another.

Topological information:

It is used to represent the construction of 3D objects, i.e. the relationship between elements of the 3D object, such as the relationship regarding layers and connections.

Then, we need to extract relevant valid data from the source file. Consider the source code of the file with the usual 3D model format STL. Assume the geometric solid represented with this file has n normal vectors of triangular facets. Thus, geometric solids in this file have 3n vertexes.

facet normal a1 b1 c1 // a1,b1,c1 denote the components of the normal-vector triangular facet along the coordinate axis

```
outer loop          //Define the start of the format for the first triangular facet
vertex x1 y1 z1     //Coordinate of the first vertex in the triangular facet
vertex x2 y2 z2     //Coordinate of the second vertex
vertex x3 y3 z3     //Coordinate of the third vertex
endloop             //The three coordinates determine the geometric location of this triangular facet
endfacet            // The first triangular facet has been defined
……                  //Proceed to define the format of the next triangular facet
endsolid filenamestl // The file ends [5]
```

(1) Extraction of the information on point, edge and surface

Let XX and XX denote the coordinates of the three vertexes of the nth triangular facet. Let XX denote the set of all vertexes of this geometric solid, and also let XX denote the set of all normal vectors of this geometric solid. From this set and the observation that there are n triangular facts, it can be inferred that the set XX contains all edges of this geometric solid.

Hence, we obtain all valid data from the source file.

What we discuss above is an example with STL as the format of the source code file. In order to support new format, we need to develop coding rules for the new format, extract the key information on elements of the 3Dmodel, and incorporate the obtained information into the sets V, P and E. In this way, we can import all information of the geometric solid. Take the AMF file as an example. The standard AMF file contains three formats of data: the data on triangular facet, the data on normal vector and the data on the vector. Because the point, edge and surface elements can be transformed to one another, we can form the foundation for online display of this file by extracting the desired information on geometric solid from file transformation.

(2) Output of the information on point, edge and surface

We need to construct new file after extracting all of the data, and generate the code that enables online display of the model by following the syntax rule of Three.js. According to the

sets V, P and E that are contained in the 3D file, we can define the file of the 3D model using the following set of data in Three.js.

where each tuple denotes the coordinates of a vertex.

Each tuple denotes the normal vector of the nth triangular facet. The data on XX and XX can be extracted from the STL source file.

Hence, the generic code developed using Three.js for the STL file is as follows.

```
 glColor3ub(192,192.192); //construct a 3D model that entirely consists of triangular facets
geBegin(GL_TRIANGLES);
//num denotes the number of triangles in the grid
//a[n],b[n],c[n] denote the components of the triangular facet's normal vector along the x, y and z axes
For(int I = 0; I < n; i++){
        Traiangle arr; //array of the triangle's vertexes
        Float t[3];         //the normal vector of the triangular facet t[n]
        t[0] = a[n];
        t[1] = b[n];
        t[2] = c[n];        //assign values to the three coordinates of the normal vector
        glNormal3fv(t); //define the normal vector
        glVertex3fv(arr.x1); //define the first vertex
        glVertex3fv(arr.y1); //define the second vertex
        glVertex3fv(arr.z1); //define the third vertex
        // …Proceed in the same way
       }
    glEnd(); //Construction is complete
```

For any STL file, we can extract its sets of points and vectors from the source file and define them as parameters. As shown in the code above,

a[n] is the first normal vector of the nth triangular facet,

b[n] is the second normal vector of the nth triangular facet,

c[n] is the third normal vector of the nth triangular facet.

arr.x1, arr.y1 and arr.z1 are the vertexes of the current triangular facet. The file can be displayed on the browser by substituting the above vertexes and normal vectors of the triangle into the code.

4. Design and Implementation of the Application Platform

The procedures above are only focused on transforming the core data of the 3D file into a format that can be recognized by the Web, and they cannot directly display the 3D graphics on the Web using the code. Hence, we need to render the geometric solid formed by these data and then map it onto the Web. To this end, we will construct a platform to display the 3D model.

Currently, most of the 3D models are displayed on the 2D screens. How to display the 3D model on a 2D screen is an issue that needs to be addressed. Thus, we will discuss the 3D model displaying technology.

In order to display 3D models on the Web, we first use the HTML elements to create a <CANVAS> element, and all operations are performed inside the elements. All 3D models have a height, length and width. The process of displaying the 3D model by mapping it to the 2D screen via graphics is what 3D rendering means. This process involves five basic steps. And after these five steps, a 3D model can be displayed on the Web. Fig. 1 shows the structural modules needed to display the 3D scenario on the Web. The procedures of Three.js modeling are described as follows [1].

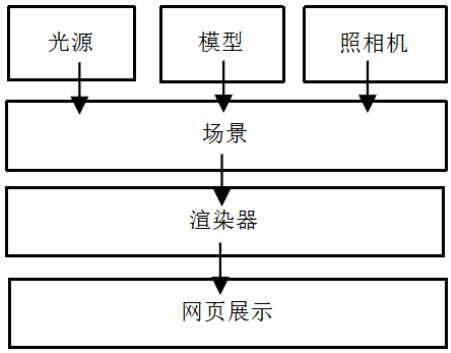


Fig. 3 Process of displaying 3D graphics on the Web

光源 Light source	模型 Model	照相机 Camera
场景 Scenario	渲染器 Renderer	网页展示 Display on the Web

(1) Scenario

The scenario can be regarded as a container or a 3D space modeled in the 2D space. It defines a 3D coordinate system and provides a frame of reference to determine the basic data of the 3D model. The 3D model which is added afterwards is added to the scenario. The effect of scenario varies with the observation distance, field of vision and angle.

(2) Camera

It defines the way in which the 3D space is projected to the 2D screen for the purpose of displaying the 3D model. In the field of graphics, it is impossible for a 3D scenario to be fully displayed in the 2D space. Although WebGL can model a 3D space, it must be output to a 2D screen. In order to address the problem of transforming the 3D scenario to the 2D space [12], we introduce the concept of "camera". The aim of this concept is to transform the 3D scenario to a 2D picture. While constructing the 3D scenario, a camera is placed at certain position to obtain a 2D projection under the viewing angle of this position.

(3) Shape and material

An object in the real-world 3D scenario usually has two properties: the geometric shape it takes and the material it is made of. The geometric shapes, like the sphere, cuboid, cylinder, and circular cone, store the information on vertexes of the geometric solid. The desired 3D model can be obtained by performing various operations (e.g., rotation, scaling, union, interaction and complement) on geometric solids. The material is a property independent of the information on the vertex A. During the modeling process, it is a set of visible properties about the surface, and it has direct influence on the brightness and visual effect of the rendered object's surface.

(4) Light source

Light source is an important factor that influences the display of 3D objects. After being rendered, the visual effect of objects made of the same material varies with light intensity and direction.

(5) Image rendering

Rendering of 3D graphics refers to the technology that enables the graphics to be viewed by users more easily by coloring in the graphics. No graphics and models can be seen on the screen without image rendering. The renderer acquires the range of rendering from

the camera and then computes the influence of light source on the object. By making use of the influence of light source, material, shape and shadow of objects, we can obtain the information on geometric properties, viewing angles, light source and texture. After the rendering process is completed, the application will obtain the 2D graphics and display the graphics on the screen.

The steps above are implemented via programming. A simple test is done on a sphere. Fig. 4 shows the model displayed on the browser.

Fig. 4 Display of 3D graphics

The discussions above illustrate the complete steps for 3D modeling on the browser. These steps can be used as the generic codes to develop a platform for the browser to dynamically create and display the 3D models for users.

To ensure efficient operation of the platform, the structure of the platform is designed as follows.

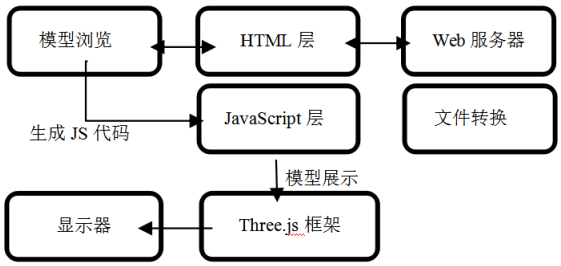


Fig. 5 Structural framework of the 3D Web system

模型浏览 Model browsing	HTML 层 HTML layer	Web 服务器 Web server	生成 JS 代码 Generate JS code	JavaScript 层 JavaScript layer
文件转换 File conversion	显示器 Screen	模型展示 Model displaying	Three.js 框架 Three.js framework	

A typical static 3D model contains information on vertexes, coordinate points, normal lines, surfaces and curves of geometric solids [12]. If the file of the 3D model is stored at the back end, the database of the web site will store a lot of information, resulting in data redundancy at the server. The proposed platform is mainly responsible for lightweight graphics viewing and file conversion, thereby highlighting the need to eliminate the burden from large data volume. In this context, by adopting the non-database design, we can relieve the pressure from the data and facilitate the reading of different files and the conversion into formats that can be recognized by JavaScript.

When the user imports relevant 3D files, this module converts the file into executable

codes. Then, the JavaScript layer of the browser uses the WebGL and Three.js frameworks to convert the files into the data that can be parsed by the browser. Next, the browser maps the data to the screen using the 3D graphics displaying technology in order to render the 3D graphics. Meanwhile, the viewing layer of the browser submits a request to the back end. The server processes the received files and provides the front end with an interface for file export. In addition to avoiding the burden of database on the server, this system architecture makes full use of the ability of the JavaScript layer to parse scripts and the ability of the front-end browser to calculate. Hence, the server enables the file conversion to be carried out more quickly and the file to be viewed much more rapidly.

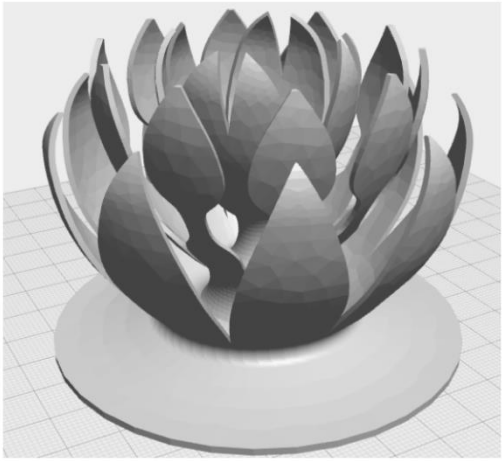


Fig. 5 An example of the displayed model

V. Results and Analysis

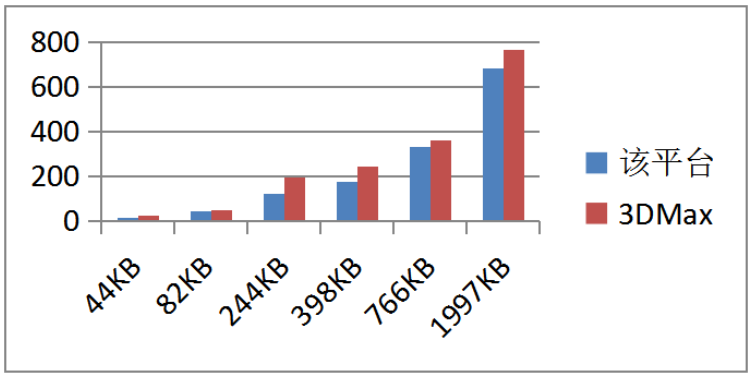
Figure 5 shows the result of the AMF file that we develop after being displayed on the platform. In order to test the efficiency of model browsing enabled by the proposed platform and the efficiency of file format conversion, Quicktest Professional is used to develop the script. The test is done on six source files of the 3D model with varying sizes (44KB, 82KB, 244KB, 398KB, 766KB, 1997KB). The amount of time consumed by the core service in 3DMax and the proposed platform is also measured.

The six files include the simplest 3D model and the large-scale 3D image. In practical applications, it may be necessary to load files of various sizes. Hence, the tests we do are typical of different situations that may occur in practice.

The amount of time consumed to display the model and convert the file is as follows.

Table 2 Time needed for the AMF file to be read, rendered and displayed on the screen (ms)

该平台: the proposed platform



The red and blue blocks denote the amount of time needed for the STL file to be opened in 3DMax and in the proposed platform, respectively. All of the six test files can be read and displayed within 1s in the proposed platform. Compared with 3DMax, the amount of time needed by the AMF file to be read, rendered and displayed on the screen increases by about 10%.

The criterion for the success of the proposed application is the ability to display the 3D model online. Experimental results show that the proposed application is able to render high-quality models quickly. Accuracy of the model is shown in Figs. 4 and 5. As for the amf format, the proposed system is able to render the model within 1s and display all models less than 2000 KB, resulting in improved model browsing efficiency.

VI. Conclusions

With the formulation of national policies to stimulate the development of Internet +, the users prefer the system which can process 3D files using the same interface and display the 3D files concisely, intuitively and easily. This type of system plays a big role in industrial transformation. Currently, Web 3D file formats have not been standardized, and a broad variety of new technologies and methods spring up. Although some issues about 3D image processing have not been addressed, some technologies have achieved remarkable results [14].

In order to address the problems associated with the use of 3D models in daily life, we propose a novel idea to enable online viewing of files in different formats, and describe the software development principles and technological methods. The 3D model displaying functions are optimized to enhance the performance of the proposed application and improve user experience. Our work yields insights into the development of 3D graphics applications in the future.

References

[1] Mr Doob:Three.js documentation http://threejs.org/docs/index.html [EB/OL] Mr Doob 2016.4.

[2] Jonathan D. Hiller, STL 2.0: A proposal for a universal multi-material additive manufacturing file format. [J]. Solid Freeform Fabrication Symposium (SFF'09), Aug 3-5 2009, Austin, TX, USA.

[3] Khronos: WebGL-OpenGL ES 2.0 for the Web. [EB/OL]. https://www.khronos.org/webgl/ . Khoronos.org, 2016.4.3.

[4] Wikipedia: WebGL(Web Graphics Library).[EB/OL]. https://en.wikipedia.org/wiki/WebGL . Wikipedia, 2016.3.31.

[5] STL (file format) https://en.wikipedia.org/wiki/STL_(file_format) [EB/OL] Wikipedia, 2016.3.31.

[6] Mr Doob: Three. jshttps://github.com/mrdoob/three.js[EB/OL]Mr Doob 2016.4.2.

[7] ZHOU Li, SU Hong-gen. Research on general file formats and algorithms of 3-D models and their implementation with OpenGL [J]. Computer Engineering and Design. 2009, 02: 433-436+439.

[8] Additive Manufacturing File Format https://en.wikipedia.org/wiki/Additive_Manufacturing_File_Format [EB/OL] Wikipedia, 2016.3

[9] BIAN Min-jie, GAO Jue, GAO Hong-hao, XU Jie-pin. Research and Application of Web3D Visualization Technology [J]. Computer Technology and Development. 2015，06：141-144.

[10] ZHENG Hua, SU Jingfang. Web-oriented 3D model generation and processing technology [J]. ,Modern Electronics Technique, 2015，24：83-86.

[11] Jiang Yanping, Xia Wangsheng, Huang Xinyuan. Comparison of Several of Web3D Techniques [J]. Journal of North China University of Technology, 2003，01：21-25.

[12] HE Long, DU Peng. WebGL-based 3D Visualization [J]. Science & Technology Information, 2015，30：23+25.

[13] LIU Yunzeng. 3D Technology for the Web: Web3D[J]. Computer and Network，2003，10：32-35.

[14] HAN Yi．Latest Development on Web3D and Web 3D Visualization——Take WebGL and O3D as the Example [J]．Science and Technology Square，2010，05：81-86．

[15] WANG Lei; GAO Jue; JIN Ye; XU Hua-hu. Research on Three-dimensional Model Display Based on Web3D without Plug-in [J]．Computer Technology and Development，2015，04：217-220．

[16] HUANG Jian．Design and Realization of Web3D Virtual Exhibition Hall Oriented Universality [D]．University of Electronic Science and Technology of China，2012．

[17] ZHAO Kong-yang; ZHU Jun; YIN Ling-zhi. Research on Network Roaming of Virtual Scene Based on WebGL [J]. Geomatics & Spatial Information Technology，2015，12：40-43．

[18] Liu Bo, Zhang Hongbin, Wang Jing. Compression and Online Browsing of 3D Graphic Data [J]. ,Computer Engineering and Applications, 2004, 28: 44-45+211.

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